<u>Economics and Society Coursework</u>

Part 1:

1) Prompt used:

   You have been given a document (the attached file).

   Use only information contained in that document. Do not use memory or external websites. If the document lacks information needed for a confident statement, say so and label it as an information gap.

   Produce a 300-word summary (approximately 300 words) written on the document.

   The summary must:
   - Open with one sentence stating the main change(s) introduced by the VAT policy on private schools.
   - Concisely describe the likely benefits and costs.
   - Explain short-run vs long-run effects where relevant.

   Overall, the summary should cover the main changes introduced by the policy and its likely costs and benefits.

   Do not speculate beyond what the document supports. Ensure the writing is concise, structured, and coherent, and remains focused on costs, benefits, and economic mechanisms.

2) Output from the AI:

   The policy would add VAT at the main rate to private school fees, removing related tax reliefs while exempting local-authority-funded specialist placements; revenues would be allocated to increase state school spending. The clear fiscal benefit is substantial: the report estimates around £1.6 billion of extra tax receipts before behavioural effects, reflecting a net VAT yield after input deductions and other adjustments. This could fund roughly a 2% uplift in day-to-day state spending in England targeted at disadvantaged pupils. Distributional effects are largely progressive: private schooling is concentrated among high-income families, so taxing fees shifts costs toward better-off households and

can modestly reduce inequality. Costs include higher out-of-pocket payments for purchasers, additional administrative complexity, scope for avoidance, and possible small reductions in labour supply where fees form a material share of household income. Behavioural responses are central: historical evidence points to weak price responsiveness, so short-run changes in private attendance are likely to be small, yet medium to long-run effects may be larger. The report judges that a 15% effective VAT could reduce private attendance by about 3–7%, equivalent to roughly 20,000–40,000 pupils, which would increase state-sector costs. Estimated marginal cost per transferring pupil is lower than average spending; central assumptions imply extra school costs of roughly £100–300 million per year. Moreover, even if private attendance falls, aggregate VAT revenues need not decline because spending may reallocate to other VATable goods and services. Key uncertainties remain: elasticity, avoidance, measurement of costs, and heterogeneity across schools; clear legal definitions, strong monitoring and contingency planning are therefore essential design features to limit unintended consequences and ensure revenues reach intended pupils. Policy evaluation should track enrolment flows, revenue timing and distributional impacts over short and long horizons, periods too.

3) Comments on AI's output

The key points of the IFS report are accurately summarised by the AI. According to the IFS (IFS, 2023, p.3; Table 2, p.15), the headline fiscal claim that roughly £1.6 billion in additional VAT receipts before behavioural effects is consistent with the idea that this could finance a minor increase in regular school spending. Given that private education is concentrated among higher-income households, it is reasonable to say that the policy is "largely progressive" (IFS, 2023, p. 9). However, this requires a disclaimer from lectures that people's reactions can undermine the apparent progressivity from earmarked spending because VAT incidence is dependent on behavioural responses (Lecture 2, slide 26).

Also, the AI accurately identifies important uncertainties. The IFS's main scenario, which is explicitly scenario-based and depends on elasticities and marginal-cost assumptions, suggests that a 15% effective VAT could reduce private attendance by roughly 3–7% (€20k–40k students) and imply additional public costs in the low hundreds of millions (IFS, 2023, p.21-24; p.24-27). The report's own concerns are mirrored in warnings about avoidance, administrative complexity, and the requirement for clear legal design (IFS, 2023, p.17-19).

4) Edited summary

The policy would increase state school spending by applying VAT to private school fees (removing existing exemptions) and exempting specialist placements funded by local authorities. Before behavioural effects, the Institute

for Fiscal Studies estimates an additional £1.6 billion in tax revenues (IFS, 2023, p.3; Table 2, p.15). Even though VAT is a regressive consumption tax, allocating those revenues to support underprivileged students is likely to result in a progressive outcome because private education is concentrated among wealthier households (IFS, 2023, p.3; Lecture 2, slide 26). Therefore, the main advantages are evident: an increase in fiscal revenue and the possibility of redistribution through higher public education spending aimed at those in need.

If the legal definition of private schooling is not strictly enforced, the main costs include higher out-of-pocket expenses for families who continue to use private provision, increased administrative complexity for regulators and schools, and opportunities for tax evasion (IFS, 2023, p.17-19). A 15% effective VAT could reduce private attendance over the medium to long term by about 3-7 percent (roughly 20,000-40,000 pupils), which would in turn create costs for the state sector. However, the IFS judges that short run demand is relatively inelastic, so immediate falls in private attendance are likely to be small (IFS, 2023, p.21-24).

According to the report's main assumptions, the package still results in a net public finance gain even though the additional state cost is small, roughly £100-300 million per year (IFS, 2023, p. 24-25). Several uncertainties that affect incidence and revenue timing, including the true elasticity of demand, the potential for avoidance, and heterogeneity among schools (IFS, 2023, p.24-27; Lecture 2, slides 21-26). In order to ensure that revenues deliver the intended equity gains without unintended consequences, the IFS advises careful legal definition, monitoring, and contingency planning for policy design (IFS, 2023, p.27).

Part 2:

1) $U_priv = P_{\mathrm{i}} + \frac{10\sqrt{F}}{y} - (1+\tau)t$

Substitute F = yt

$$\mathrm{U_priv} = P_{\mathrm{i}} + \frac{10\sqrt{yt}}{y} - (1+\tau)t$$

Simplify the equation

$$U_priv = P_{\mathrm{i}} + \frac{10\sqrt{t}}{\sqrt{y}} - (1+\tau)t, \qquad i \in \{L, M, H\}$$

P is the baseline private benefit. t is tuition, τ is the VAT rate and y is private enrolment.

i ∈ {L, M, H}, where L represents low-income family, M the middle-income family, and H the high-income family.

The private payoff has three parts. P_{i} represents why some families choose private schooling regardless of price based on non-price factors like status, school fit, reputation of school. In practice this means wealthier families have a higher P_{i} and are more resilient to price rises. The term $\frac{10\sqrt{t}}{\sqrt{y}}$ shows the per pupil private school benefit rises with $\sqrt{t}$ (tuition) and falls with $\sqrt{y}$ (number of private school pupils). Finally, $(1+\tau)t$ is the full monetary cost of the tuition, which is tuition plus VAT. Increasing τ raises out-of-pocket charges directly and is a factor that discourages private schooling up take.

This links to the costs and benefits mentioned as VAT revenue comes from the $(1+\tau)t$ term and can fund public schooling (fiscal benefit, IFS, 2023, p.3), in turn, distributional consequences result from the concentration of private students (those with greater Pi, IFS, 2023, p.9). Behavioural responses are driven by price increases, but they are moderated by quality feedback. Leaving a private school, for instance, reduces y, which increases per-pupil private quality and may partially offset departures (explains small, short-run elasticities but is more ambiguous to medium-run effects, IFS, 2023, p.21-24; Lecture 9, 2024, slides 7-9). Finally, the theoretical VAT yield may be lower in practice due to collection costs and avoidance risk, so careful legislative design and monitoring are crucial (IFS, 2023, p. 17-19; Lecture 9, 2024, slide 26).

2)

T = private tuition; τ = VAT rate: y = number of private pupils (y = 3 - x); x = number in public.

P_{i} = baseline private benefit for family $i \in \{L, M, H\}$with $0 \leq P_L < P_M < P_H$

Total private fees $F = yt$. Total VAT revenue $T = \tau F = \tau yt$

Public payoff equation:

$$Ui = \frac{S}{x} + 5 \times \frac{T}{x}$$

Private payoff equation:

$$U_priv = P_{\mathrm{i}} + \frac{10\sqrt{t}}{\sqrt{y}} - (1+\tau)t$$

When x = 1 and y = 2 (only L attends public, M & H attend private)

Public:

U(x = 1) = S +5T Sub in $T = \tau F = \tau y t$ and make y = 2

= S + ($2\tau t$)

= S + 10 τt

Private:

U_priv(y = 2) = $\boldsymbol{P_{\mathrm{i}} + \frac{10\sqrt{t}}{\sqrt{2}} - (1+\tau)t}$

Low-income family (L) prefers public:

$$\mathbf{S + 10\,\tau t} \;\geq\; \boldsymbol{P_L + \frac{10\sqrt{t}}{\sqrt{y}} - (1+\tau)t}$$

Middle income family (M) prefers private:

If M switched to public it would be x = 2 and y = 1

That would be $\frac{S}{2} + \frac{5}{2}\,\tau t$

$$\mathrm{P_M} + \boldsymbol{\frac{10\sqrt{t}}{\sqrt{2}} - (1+\tau)t} \;\geq\; \boldsymbol{\frac{S}{2} + \frac{5}{2}\,\tau t}$$

High-income family prefers private: If it was to switch to public the private payoff would be same as M assuming all other families don't change their decision.

$$\mathrm{P_H} + \boldsymbol{\frac{10\sqrt{t}}{\sqrt{2}} - (1+\tau)t} \;\geq\; \boldsymbol{\frac{S}{2} + \frac{5}{2}\,\tau t}$$

<u>The numerical figures to satisfy these conditions are below.</u>

$t=1,\ \tau=0.20,\ P_L=1,\ P_M=5,\ P_H=8,\ S=10$

For P_L

(left) $S + 10\ \tau t = (10) + 10\ (0.2)(1) = 12$

(Right) $1 + 7.07 - 1.2 = 6.87$

$\mathbf{12 \geq 6.87}$ **so conditions are satisfied**

For P_M

(Left) $5 + 7.07 - 1.2 = 10.87$

(Right) $\frac{S}{2} + \frac{5}{2}\ \tau t = 5 + 0.5 = 6.5$

$\mathbf{10.87 \geq 5.5}$ **so conditions are satisfied**

For P_H

(Left) $8 + 7.07 - 1.2 = 13.87$

(Right) $=5.5$ (Same as P_M)

$\mathbf{13.87 \geq 5.5}$ **so conditions are satisfied**

3)

X=2 y=1 as now 2 families choose public

Based on last question I know when x=2 **public payoff** is:

$$U(x = 2) = \frac{S}{2} + \frac{5}{2}\ \tau t$$

Private payoff when y=1:

$$U_priv = P_{\mathrm{i}} + \frac{10\sqrt{t}}{\sqrt{y}} - (1+\tau)t$$

$$U_priv\ (y=1) = P_{\mathrm{i}} + 10\sqrt{t} - (1+\tau)t$$

Low-income family prefers public so:

$$\frac{S}{2} + \frac{5}{2}\ \tau t\ \geq P_L\ +\ \frac{\mathbf{10\sqrt{t}}}{\sqrt{\mathbf{2}}} - \boldsymbol{(1+\tau)t}$$

Middle income prefers public. For the private payoff, if M changes to private the payoff would be x=1, y=2 (same as private payoff in previous question):

$$\frac{S}{2} + \frac{5}{2}\ \tau t\ \geq \mathbf{P_M} + \frac{\mathbf{10\sqrt{t}}}{\sqrt{\mathbf{2}}} - \boldsymbol{(1+\tau)t}$$

High income family prefers private. For their public payoff, if they were to switch to public it would be at a payoff where x=3 and y=0.

If x=3 and y=0 public payoff would be $\frac{S}{3}$.

$$\mathrm{P_H} + 10\sqrt{t} - (1+\tau)t \geq \frac{S}{3}.$$

Numerical values example

t=1 , τ=0.20 , PL=1 , PM=5 , PH=12 , S=22

$$\frac{10\sqrt{t}}{\sqrt{2}} \approx 7.0711.$$

$$(1+\tau)t = 1.2$$

$$\frac{5}{2}\tau t = 0.5.$$

$$\frac{S}{2} + \frac{5}{2}\tau t = 11 + 0.5 = 11.5.$$

$\frac{S}{3} \approx 7.333.$

<u>Low income</u>

(Left) $\frac{S}{2} + \frac{5}{2}\tau t = 11 + 0.5 = 11.5.$

(Right) $PL \ + \frac{\mathbf{10\sqrt{t}}}{\sqrt{\mathbf{2}}} - (\mathbf{1} + \boldsymbol{\tau})\boldsymbol{t} \mathbf{=} \mathbf{1} \mathbf{+} 7.0711 - 1.2 = 6.8711$

$\mathbf{11.5} \ \geq \mathbf{6.8711}$ **so, conditions satisfied**

<u>Middle income</u>

(Left) $\frac{S}{2} + \frac{5}{2}\tau t = 11 + 0.5 = 11.5$

(Right) $\mathbf{5} \mathbf{+} 7.0711 - 1.2 = 10.8711$

$\mathbf{11.5} \ \geq \mathbf{10.8711}$ **so, conditions satisfied**

<u>High income</u>

(Left) PH $+10\sqrt{t} - (1 + \tau)t = 12 + 10 - 1.2 = 20.8$

(Right) $\frac{S}{3} \approx 7.333$

$\mathbf{20.8} \ \geq \mathbf{7.333}$ **so, conditions satisfied**

4)

Private payoff

$$U_priv = P_{\mathrm{i}} + \frac{10\sqrt{t}}{\sqrt{y}} - (1 + \tau)t$$

Public payoff

$$U(x) = \frac{S}{x} + 5 \times \frac{T}{x} = \frac{S}{x} + 5 \times \frac{\boldsymbol{\tau y t}}{x}$$

Scenario 1: $\boldsymbol{x = 1, y = 2}$.

Scenario 2: $\boldsymbol{x = 2, y = 1}$.

Effects on the likelihood of Scenario 1:

- S : A small decrease in S increases the likelihood of Scenario 1. This is because lowering the baseline public benefit makes public schools less appealing to middle-class and upper-class families, so M and H are less likely to transfer from private to public education.
- τ (VAT rate): In most situations, a decrease in τ increases the likelihood of Scenario 1. This is because lower VAT lowers both the additional public funding and the price penalty on private fees, public becomes less appealing to M/H while private becomes comparatively more affordable, assisting M and H in staying private. However, if τ drops too much, L may no longer find the public appealing.
- t (tuition): Increase t can make Scenario 1 more likely *if* the quality gain dominates the price rise. This is because the private school quality term 10 $\frac{\boldsymbol{10\sqrt{t}}}{\boldsymbol{\sqrt{y}}}$ is increased by higher tuition. It increases the out-of-pocket expense but also makes private more appealing to M and H. Therefore, raising t only benefits Scenario 1 when the additional perceived private benefit exceeds the additional expense for M and H.
- Lowering P_L makes scenario 1 more likely. This is because reducing P_L reduces their private payoff so they are more likely to stay public.
- Increase in P_M or P_H makes scenario 1 more likely. This is because a larger P_M or P_H raises those family's private payoffs so they prefer to remain private, which is exactly what Scenario 1 requires.

5)

Increasing the value-added tax (VAT) on private school fees can increase overall social welfare if school choices remain unchanged. This is because it transfers resources from private families to the public school system in a way that benefits more people. Families attending private schools pay more out-of-pocket, making them less fortunate on an individual basis, but the VAT they pay is combined and used to boost public school funding. Because all students in public schools receive this additional funding, the overall benefit to public school families may outweigh the overall loss to private school families. In this instance, VAT serves as a tool for redistribution, moving funds from households with generally higher incomes to better public education. However, behaviour determines this beneficial welfare effect.

The welfare advantage may be reduced or even eliminated if a higher VAT causes many families to transfer their children from private to public schools, making the public system more congested and costly to operate. If VAT is evaded or is expensive to collect, welfare may also decline. In summary, VAT can lower welfare when behavioural and administrative costs are high, but it can raise welfare when reactions are low and revenue is used efficiently.

6)

In theory, the imposition of VAT on private school fees is justified, but it is risky. Our theoretical model illustrates the immediate effect. While the tax revenue can be combined to increase funding for public schools, adding VAT increases the amount of money families must pay for private education, which lowers private families' disposable income. The redistribution can raise overall welfare if school choices do not significantly change because public gains (spread among numerous students) may exceed private losses. This is consistent with the lecture discussion that public funding can address inequality and capture wider social returns to education (lecture 9, 2023, p.8).

Evidence, however, alerts us to trade-offs. VAT raises the cost of private education and may force marginal (usually middle-class) families into public schools, increasing crowding, potentially lowering average quality in both sectors through composition effects, and raising marginal public costs. Peer effects and the concentration of wealthier students in private schools are highlighted in empirical slides, therefore, fee taxation is progressive but may result in disruptive changes in enrolment and outcomes (lecture 9, 2023, pp.14-15, 21-24).

It's evident who wins and who loses. Students attending public schools, many of whom have lower incomes, benefit from increased funding and possibly less segregation. Families who pay more for private schools are the losers, and if enrolment declines, private quality may suffer. The wealthiest households are least impacted because non-financial factors (status, networks) keep them enrolled in private schools, while middle-class households are most vulnerable to being pushed across the divide.

Designing policies is essential with reforms that could include a modest VAT with robust anti-avoidance measures, targeted assistance for underprivileged students , and phased implementation to allow public capacity to grow to generate revenue while minimising harm. To maintain choice while lowering inequality, lecture materials emphasise the importance of complementarities between public funding and targeted policy (lecture 9, 2023, p.15). In general, VAT on private fees can be included in a package of fair policies, but only if it is applied carefully and in conjunction with safeguards.

Part 3:

1)

Predictions:

- When the VAT rate on private fees (τ) rises, the number of children attending private schools fall as people leave.
- A higher τ raises VAT revenue from private fees and increases public funding.
- If the number of people switching from private to public is small, then higher τ improves average public outcomes as there is more money in public leading to increased resources.
- If the number of people switching from private to public is large, then a higher τ would lower public outcomes as there would be a crowding in public schools leading to a stress on resources.
- Higher tuition fees (t) improve private school quality per pupil but also increases the price. The net effect on private school enrolment all depends on the price elasticity.
- VAT is a progressive tax, and so higher-income households take most of the tax burden as private schools are mainly used by richer families. Middle-income households are most like to switch when tax rises.

2)

When the VAT rate on private fees (τ) rises, the number of children attending private schools fall as people leave.

Variables to measure:

- Dependent variable: private enrolment (count or share of pupils in private schools).
- Independent variable: VAT policy indicator (timing of VAT introduction / changes in τ).

Data sources:

Independent Schools Council (ISC) Annual Census: counts of pupils in independent schools. Use the ISC pupil counts by school and year.
_isc_census_2025_29may25.pdf

Department for Education (DfE) : total pupils by area / state vs independent split; use to form private share which would be ISC_private_pupils / total_pupils_by_area.

Schools, pupils and their characteristics, Academic year 2024/25 - Explore education statistics - GOV.UK

HMRC / Government publications for VAT policy/timing and receipts

HMRC tax receipts and National Insurance contributions for the UK - GOV.UK

3)

- Theory - When the VAT rate on private fees (τ) rises, the number of children attending private schools fall as people leave.

$$Private_{at} = \alpha + \beta VAT_t + \gamma X_{at} + \mu_a + \lambda_t + \varepsilon_{at}$$

$Private_{at}$: Share of pupils attending private schools in area in α in year t. This is constructed form ISC private pupil counts divided by total pupils from DfE data.

VATt: A policy variable capturing the VAT rate on private school fees. This reflects the parameter T in the model.

Xat: A vector of control variables such as average household income and population size. These control for other factors affecting school choice.

μ_a: Area fixed affects. These control for time invariant differences.

Λ_t: Year fixed effects. This captures macro-economic conditions e.g. nationwide shocks.

E_{at}: This is the error term.

The main coefficient of interest is β.

This is the hypothesis to test:

Null hypothesis: $\beta = 0$

Alternate hypothesis: $\beta < 0$. This would suggest an increase in VAT reduces private school enrolment.

If β is negative and statistically significant, it supports the theory's prediction that higher VAT pushes families away from private schools.

4)

This model may suffer from endogeneity issues for a few reasons listed below.

- Omitted variables: Private enrolment and VAT policy discussions may be impacted by unobserved factors like shifts in public school reforms or attitudes towards private education. $VATt$ will be correlated with the error term if these are not fully captured by controls.
- Policy endogeneity (reverse causality): Due to the high or increasing use of private schools, the government may implement VAT. Prices and preferences determine private enrolment in the model and if enrolment trends impact policy, causality is reciprocal.
- General equilibrium effects: VAT has an impact on public funding and crowding, as was demonstrated in previous sections. The interpretation of β may be complicated by the indirect influence of these feedback effects on school choices.

Potential fix:

Difference-in-differences (DiD): Examine how private enrolment changed before and after the implementation of the VAT in comparison to a control group (such as age groups or less exposed regions). This is practical and takes advantage of the policy's timing.

Part 4:

Note: In this answer video 1 refers to first link on dangers of VAT and video 2 refers to the second link with the LBC video discussion from host and caller.

The videos highlight significant, pressing issues: VAT raises the cost of private education, which drives middle-class families out of the market and puts more strain on public schools (Video 1). It may be viewed as a "jealousy tax" that damages private schools and puts them at risk of closure (Video 2). The fundamental price-driven mechanism is captured by our theoretical model as higher VAT increases the out-of-pocket expense of private education and, ceteris paribus, decreases private enrolment. The model also accounts for distributional effects as middle-class families are more likely to transfer to public schools, while wealthier households with stronger non-price preferences stay private. This directly relates to the main concerns raised in the videos regarding middle-class switching and overcrowding in public schools.

Nevertheless, the model does not include a few of the arguments in the videos. In contrast to the three-family model, the speakers emphasise system capacity (teacher shortages, strikes), changing supply responses (private school closures), macro spillovers (lower household spending), and behavioural responses. The model should be expanded to include households across income percentiles (rather than just three types), permit private schools to modify capacity and price, add a public-school capacity constraint and per-pupil marginal cost, and model dynamic responses in order to analyse these. These changes enable us to examine crowding, closure risk, and short versus long-term impacts on costs and quality.

The ISC Annual Census (school level pupil counts), DfE data (total pupils by area to form private shares), and HMRC records for VAT timing can all be used to empirically test the key prediction, "$\uparrow \tau(\mathrm{VAT\ rate}) \rightarrow \uparrow$ private enrolment." It is appropriate to compare pre/post trends in private shares across regions using a difference in differences or event study design that accounts for cohort size and demographics. Subgroup analysis by local income to see if middle classes respond more and looking for school closures in ISC panels are helpful robustness checks.

The main limitations still exist because policy implementation might coincide with other shocks, avoidance or fee adjustments might conceal actual effects, and closure dynamics need in-depth school studies. Use external tools or staggered implementation whenever possible and provide event plots and heterogeneity checks.

Overall, the model explains the videos price and distribution, but evaluating capacity, closures, and broader macro effects needs model extensions and more targeted empirical work.